

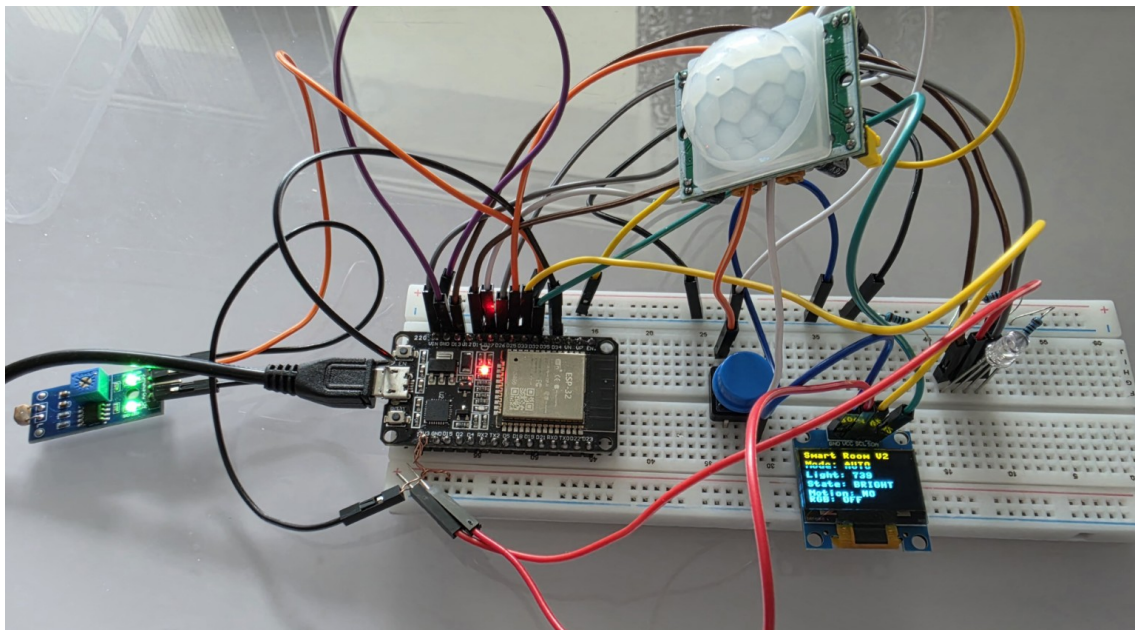
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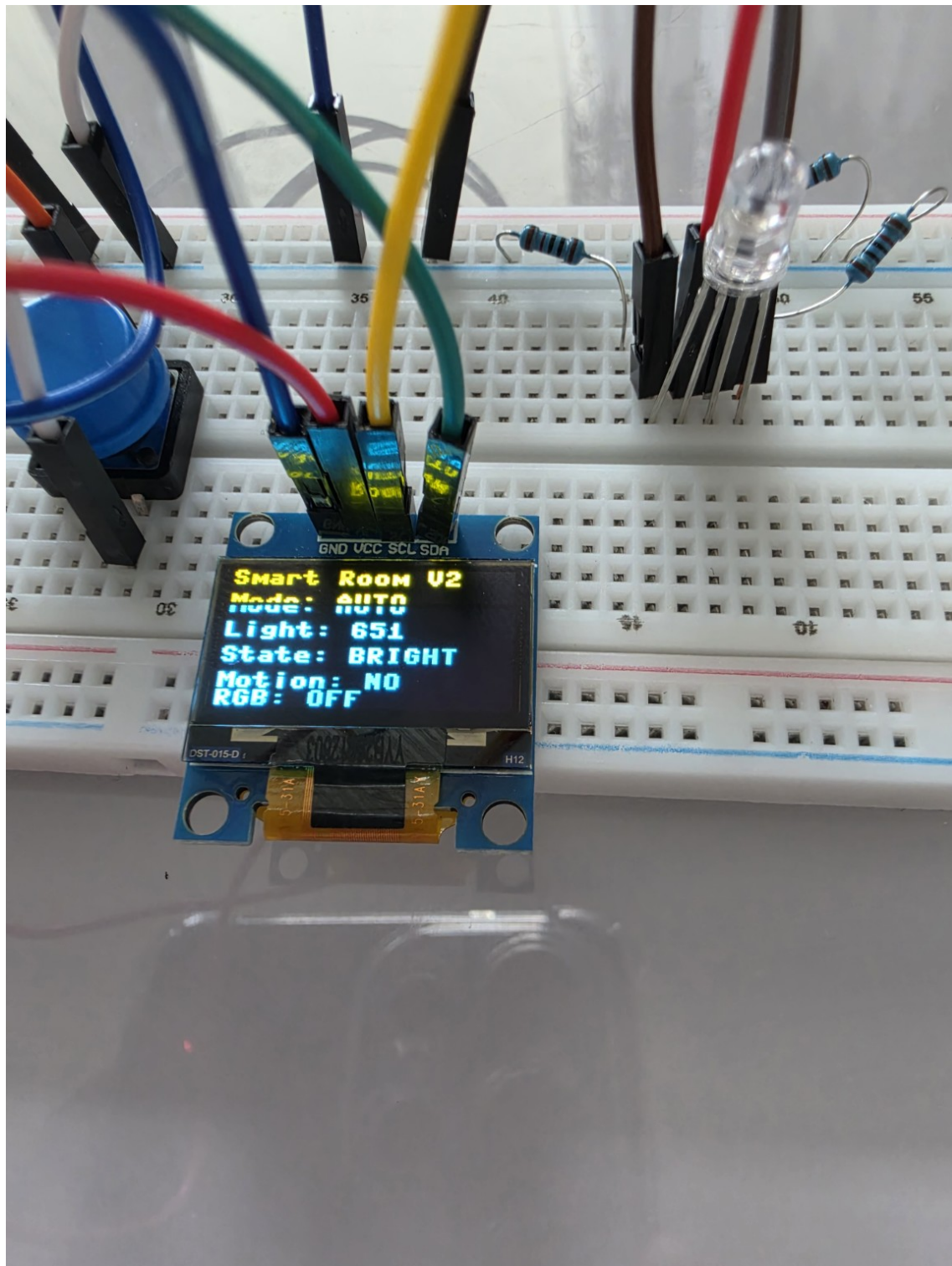
ESP32 Smart Room Controller V2 – Deutsche Dokumentation

Dieses Projekt basiert auf einem ESP32-Mikrocontroller und kombiniert mehrere intelligente Sensoren mit einem OLED-Display und einem RGB-Beleuchtungssystem. Das Ziel dieses Projekts ist die Entwicklung eines intelligenten Smart-Room-Controllers, der Umgebungslicht erkennt, Bewegungen überwacht und automatisch die RGB-Beleuchtung steuert. Das System verwendet: Einen PIR-Bewegungssensor zur Bewegungserkennung
Einen LDR-Lichtsensord zur Messung des Umgebungslichts
Eine RGB-LED mit Common-Anode-Technologie
Einen Push-Button zur manuellen Modussteuerung
Ein OLED-Display zur Echtzeit-Anzeige aller Systeminformationen
Das Projekt verfügt über drei Betriebsmodi: AUTO MODE MANUAL ON MODE MANUAL OFF MODE
Zusätzlich unterstützt das System sanfte RGB-Fade-Effekte, energieeffiziente Beleuchtungssteuerung und intelligente Umgebungsbeleuchtung.

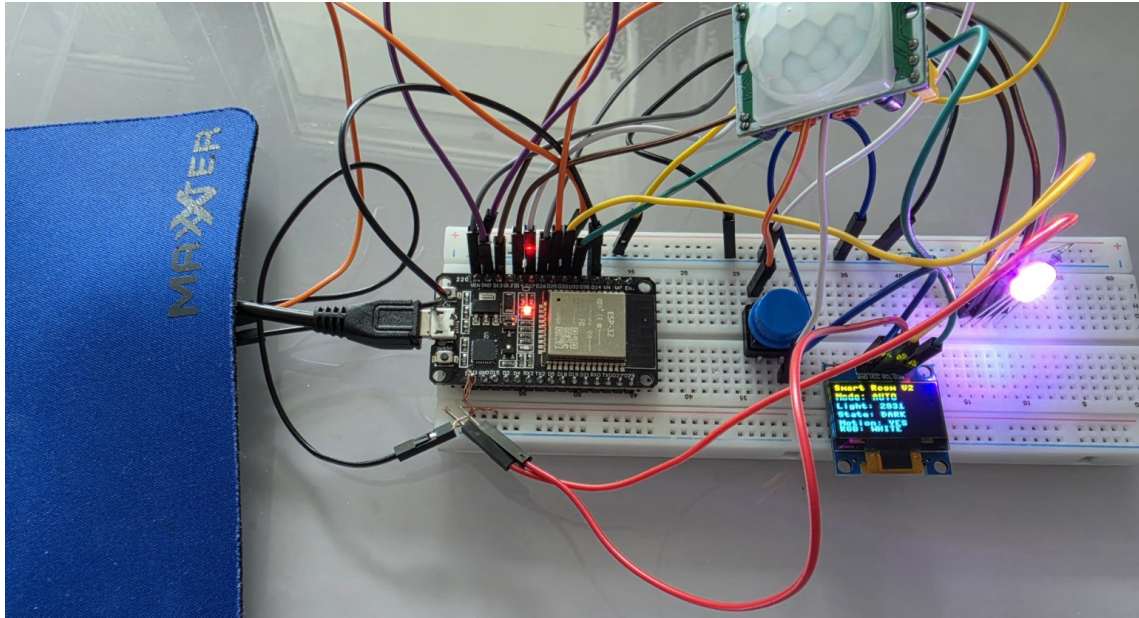
Systembilder und Live-Demonstration



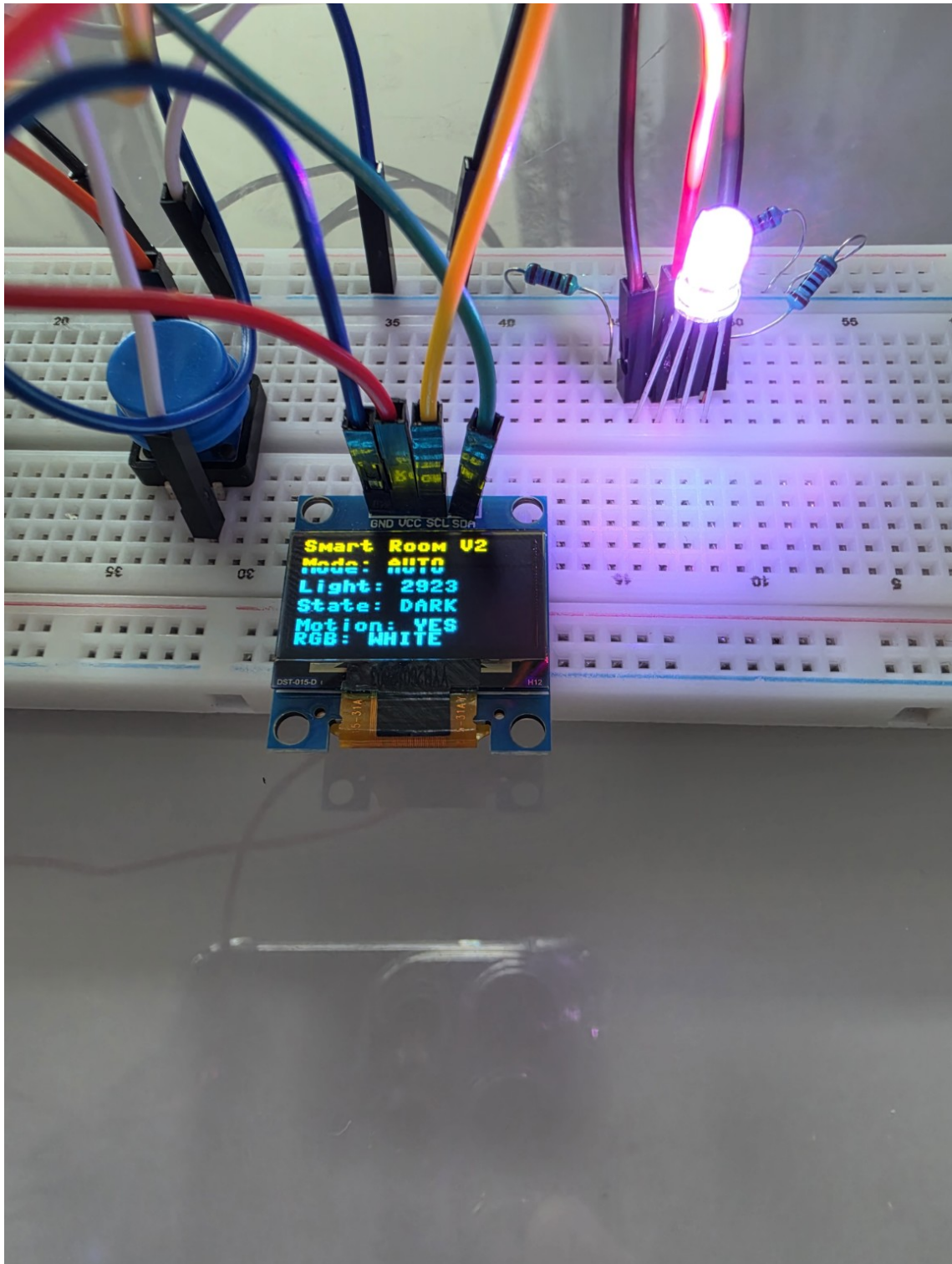
Helle Umgebung erkannt. Das System deaktiviert automatisch die RGB-Beleuchtung zur Energieeinsparung.



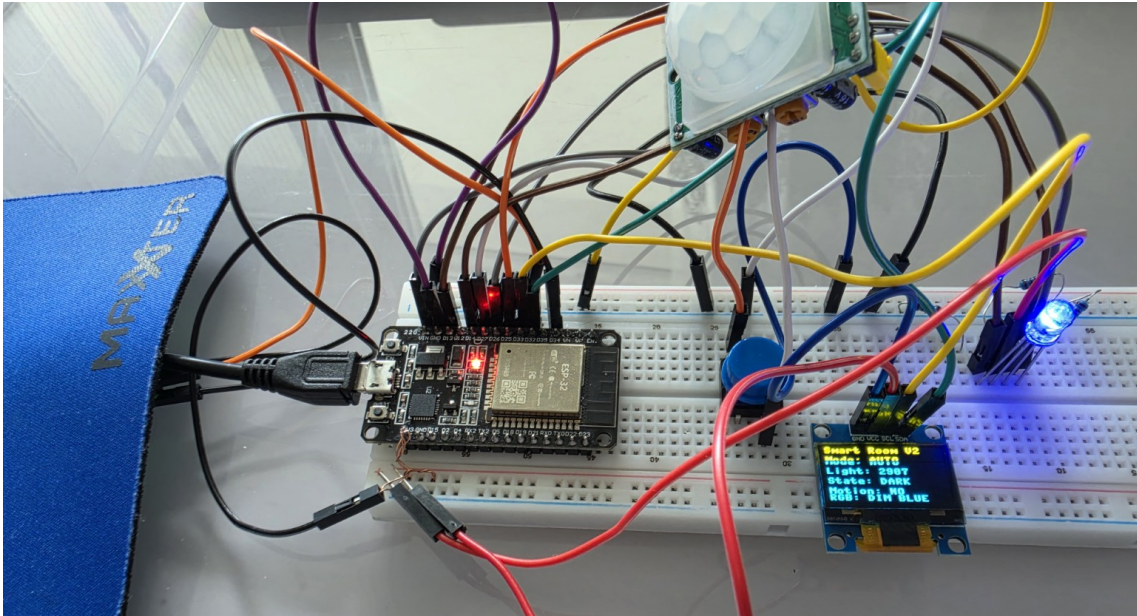
OLED-Display zeigt Live-Systemdaten in heller Umgebung an.



Dunkle Umgebung mit erkannter Bewegung. Das System aktiviert die weiße RGB-Beleuchtung.



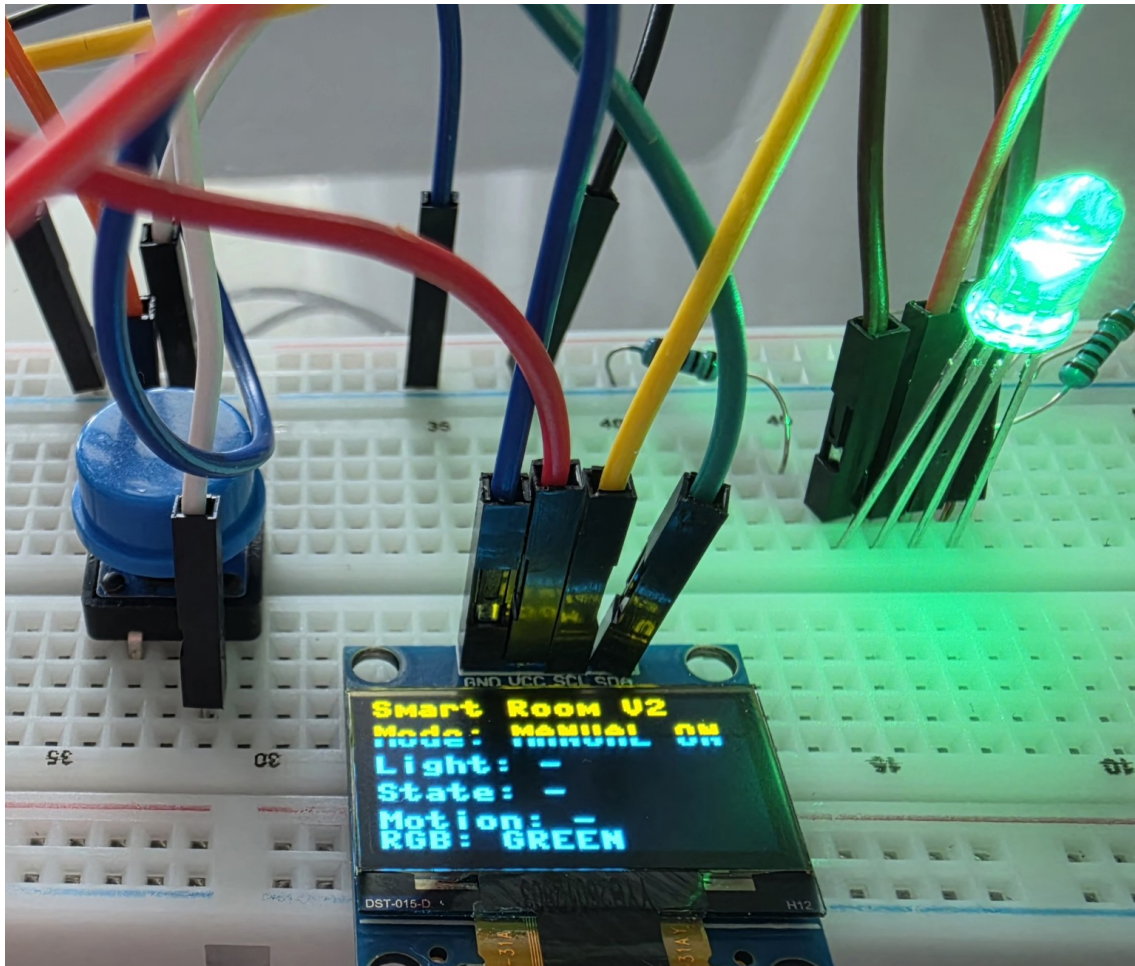
OLED-Anzeige während der Bewegungserkennung in dunkler Umgebung.



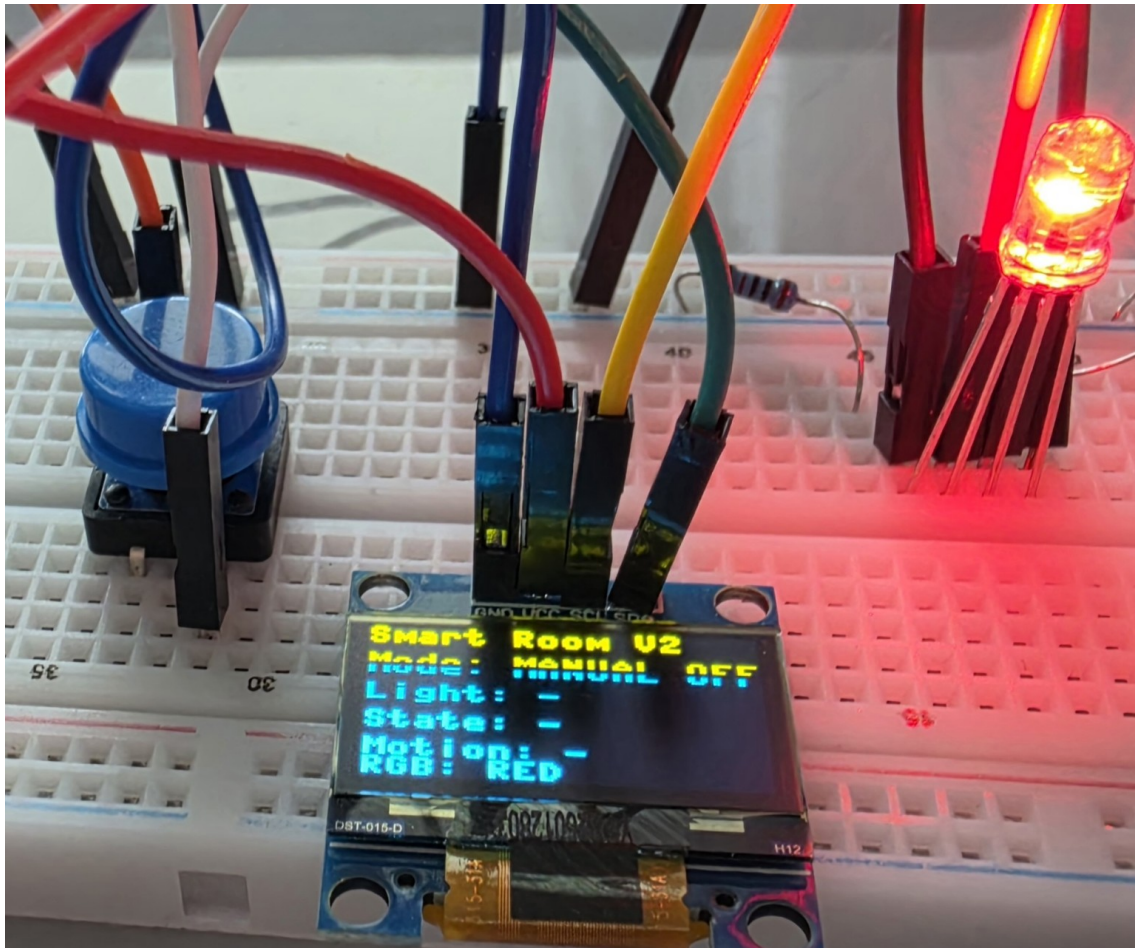
Dunkle Umgebung ohne Bewegung. Das System aktiviert ein gedimmtes blaues Ambientelicht.



OLED-Display zeigt den DIM BLUE Zustand im AUTO-Modus.



MANUAL ON MODE aktiviert. Die RGB-LED leuchtet grün.



MANUAL OFF MODE aktiviert. Die RGB-LED leuchtet rot.

Praktische Anwendungen

Dieses Projekt kann in vielen realen Anwendungen eingesetzt werden: Smart-Home-Beleuchtungssysteme
Energieeffiziente Raumsteuerung Automatische Nachtbeleuchtung IoT-basierte Gebäudeautomatisierung
Intelligente Smart-Building-Systeme Embedded- und IoT-Lernplattformen Prototypen für Smart-Energy-Systeme

ESP32 Smart Room Controller V2 – English Documentation

This project is based on an ESP32 microcontroller and combines multiple intelligent sensors with an OLED display and an RGB lighting system. The main goal of this project is to develop a professional smart room controller capable of detecting ambient light conditions, monitoring motion and automatically controlling RGB lighting behavior. The system uses: A PIR motion sensor for motion detection An LDR light sensor for ambient brightness monitoring A common anode RGB LED for intelligent status indication A push button for manual mode switching An OLED display for real-time system monitoring The project supports three operating modes: AUTO MODE MANUAL ON MODE MANUAL OFF MODE Additionally, the system supports smooth RGB fading effects, energy-efficient lighting control and smart ambient lighting functionality.

Complete MicroPython Source Code

```
from machine import Pin, ADC, PWM, I2C
import ssd1306
import time

# =====
# ESP32 Smart Room Controller V2
# OLED + RGB + PIR + LDR + Button
# =====

# RGB LED - Common Anode
red = PWM(Pin(14))
green = PWM(Pin(26))
blue = PWM(Pin(13))

red.freq(1000)
green.freq(1000)
blue.freq(1000)

# Sensors
light_sensor = ADC(Pin(34))
light_sensor.atten(ADC.ATTN_11DB)
light_sensor.width(ADC.WIDTH_12BIT)

motion_sensor = Pin(27, Pin.IN)

# Button
button = Pin(25, Pin.IN, Pin.PULL_UP)

# OLED Display
# SCL -> GPIO33
# SDA -> GPIO32
i2c = I2C(0, scl=Pin(33), sda=Pin(32), freq=100000)
oled = ssd1306.SSD1306_I2C(128, 64, i2c)

# Variables
mode = 0
last_button_state = 1
motion_timeout = 5
last_motion_time = 0
current_color = (0, 0, 0)

def update_oled(mode_text, motion_text, light_value, light_state, rgb_text):
    oled.fill(0)

    oled.text("Smart Room V2", 0, 0)
    oled.text("Mode: " + mode_text, 0, 12)
    oled.text("Light: " + str(light_value), 0, 24)
    oled.text("State: " + light_state, 0, 36)
    oled.text("Motion: " + motion_text, 0, 48)
    oled.text("RGB: " + rgb_text, 0, 56)

    oled.show()

def set_color(r, g, b):
    red.duty(1023 - r)
    green.duty(1023 - g)
    blue.duty(1023 - b)

def fade_color(start, end, steps=30, delay=0.01):
    for i in range(steps + 1):
```

```

        r = int(start[0] + (end[0] - start[0]) * i / steps)
        g = int(start[1] + (end[1] - start[1]) * i / steps)
        b = int(start[2] + (end[2] - start[2]) * i / steps)

        set_color(r, g, b)

        time.sleep(delay)

def go_to_color(target):
    global current_color

    if current_color != target:
        fade_color(current_color, target)
        current_color = target

print("ESP32 Smart Room Controller V2 Running...")

update_oled("START", "-", "-", "-", "BOOT")

while True:

    button_state = button.value()

    if button_state == 0 and last_button_state == 1:

        mode += 1

        if mode > 2:
            mode = 0

        time.sleep(0.3)

    last_button_state = button_state

    if mode == 0:

        mode_text = "AUTO"

        light_value = light_sensor.read()

        motion = motion_sensor.value()

        if light_value < 1800:

            light_state = "BRIGHT"

            motion_text = "YES" if motion == 1 else "NO"

            rgb_status = "OFF"

            go_to_color((0, 0, 0))

        else:

            light_state = "DARK"

            if motion == 1:

                last_motion_time = time.time()

                motion_text = "YES"

                rgb_status = "WHITE"

                go_to_color((500, 500, 500))

            else:

                motion_text = "NO"

                elapsed = time.time() - last_motion_time

                if elapsed < motion_timeout:

                    rgb_status = "WHITE"

                    go_to_color((500, 500, 500))

                else:

                    rgb_status = "DIM BLUE"

                    go_to_color((0, 0, 120))

    update_oled(

```

```
        mode_text,\n        motion_text,\n        light_value,\n        light_state,\n        rgb_status\n    )\n\n    elif mode == 1:\n\n        mode_text = "MANUAL ON"\n\n        motion_text = "-"\n\n        rgb_status = "GREEN"\n\n        go_to_color((0, 500, 0))\n\n        update_oled(\n            mode_text,\n            motion_text,\n            "-",\n            "-",\n            rgb_status\n        )\n\n    elif mode == 2:\n\n        mode_text = "MANUAL OFF"\n\n        motion_text = "-"\n\n        rgb_status = "RED"\n\n        go_to_color((500, 0, 0))\n\n        update_oled(\n            mode_text,\n            motion_text,\n            "-",\n            "-",\n            rgb_status\n        )\n\n    time.sleep(0.5)
```

Conclusion

The ESP32 Smart Room Controller V2 demonstrates how embedded systems, IoT technologies and smart energy concepts can be integrated into a complete practical project. The system combines automatic lighting control, motion detection, ambient light monitoring, manual override functionality and real-time OLED visualization into a professional embedded IoT prototype.

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